
A Practical Use of AI in the Church

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How Can the Church Use AI?

Proposal:

Use AI to analyze existing content, power upcoming technology, and connect people to the Church.

How Can the Church Use AI?

What is included?

- A practical starting place for AI adoption
- Free (open source) or cheap implementation - solutions possible today

What isn't?

- A comprehensive answer to the question
- Putting generative content in front of users
 - Policy is the biggest roadblock to AI adoption
- Paying other companies to control your domain



Copilot



ChatGPT

How Can the Church Use AI?

- From the context of the AG, but consider your own
- Proposed capabilities are aimed at
 - The larger organization of the AG
 - Chuches
 - The average church goer or visitor



Copilot



ChatGPT

Churches Already Produce Content

The AG is comprised of about 13000 churches

Based on research on ~600 churches in the North Texas district:

18% of churches are posting videos online

On average about 90 minutes of content per church per month is uploaded

This means for the AG in the US:

- Over 2400 churches with videos online

- Over 23000 hours of content every month



Too Much Content?

Over 130 people working full time to watch 23000 hours a month of content

AI enables analysis and use of this data in other tools

What if there was a way to utilize this content?

Are we truly able to track trends in our church services?

Could we better train pastors and meet the needs of church goers?

How do AG churches compare to other churches?

How Would This Work?

Identify Videos

Pick a church and find all the content they have available.

- Yt-dlp
- AI websearch
- Selenium scrape
- AG or district lists

Transcribe

Download captions or use AI to transcribe. Storing video files is not necessary.

- Youtube captions
- Whisper

LLM

Through a series of calls to an LLM, the transcripts can be stored in a structured form.

- Parts of service (worship, sermon, etc.)
- Local LLM or online model API

Data Lake

The data is stored in the data lake. More than just raw transcripts.

- Estimated 4gb per 1000 hours content
- More on the formatting later...

Usage

Analysis is now possible on data in the lake. Tools built can draw from the data. Existing data sources from the data lake can be used as needed.

- Analytics
- Uses for a church
- Structuring the data lake

Important Highlights

The only input is a list of churches

The data can be curated

Since we are only keeping transcripts, content is roughly 100gb

What if I don't want to see content from A or B?

Churches can be included regardless of size

Analyzing the Data - Questions

What are the most popular worship songs this month?

When a national event happens on a certain day, how many churches talk about that specific event?

How many churches preach sermon series out of the same book?

Can we create a metric to quantify biblical literacy at a church?

What are the most common sermon points when preaching from Matthew 5?

Do large churches structure their services differently than small services?

A Tool for Church Websites

The data lake can power an app that can be embedded on a church website

When a user asks a question they get an answer from the data lake

- Only returns direct quotes from services (or other data in the lake)

Churches can customize which churches they will return results for

All results from videos link to the actual video at the timestamp

Pastors learn what their people are asking

Example User Tool - Would More Users Visit?

Suppose a member of your church asks, “what is the Holy Spirit?”

The tool then returns:

1. A definition of the Holy Spirit - from your Holy Spirit service (link)
2. A discussion on the works of the holy spirit - from a panel you held for a conference your church hosted (link)
3. Other answers from other churches

Do you want to talk to pastor about this? (link to message pastor)

Church Report

What if your church received a report every 6 months?

- Trends in your district
- What questions are people from your church asking in the tool? (last slide)
- Progress towards targets from your district

What questions do you have? All questions are answered next time

Knowledge Graph

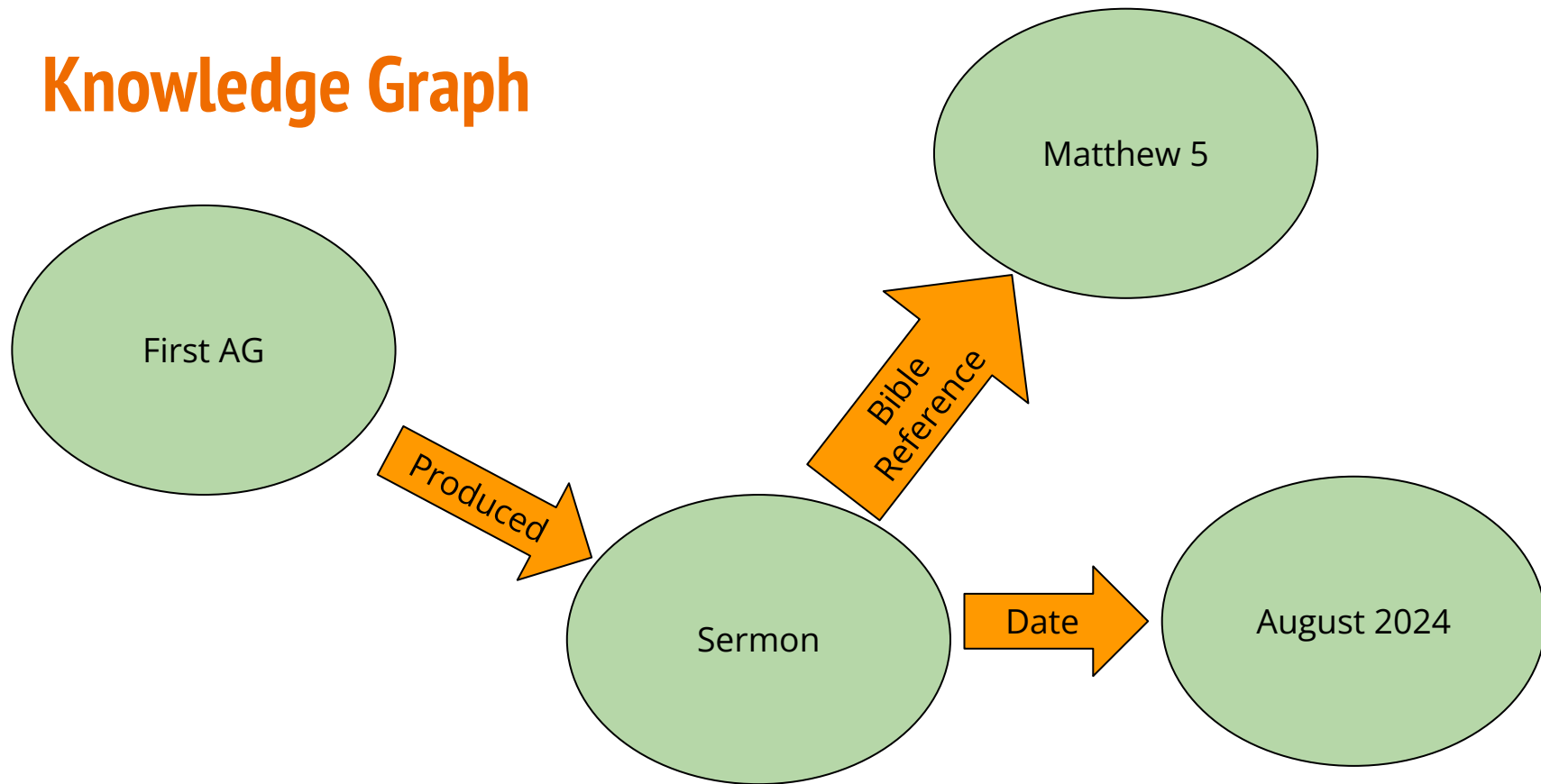
A knowledge graph stores your information along with the connections between each piece of information.

Growing this graph is a primary objective of the data lake.

Two immediate impacts of the knowledge graph:

1. You will better understand your own data including gaps, areas of saturation, and trends
2. AI Tools will thrive with this. This will be how you control their output.

Knowledge Graph



Conclusion

- This is possible today and prepares for tomorrow
- None of this is groundbreaking technology
 - Open-source and low cost
- Who should be doing this?
- Who is God calling to utilize AI? Missionaries? Students?